

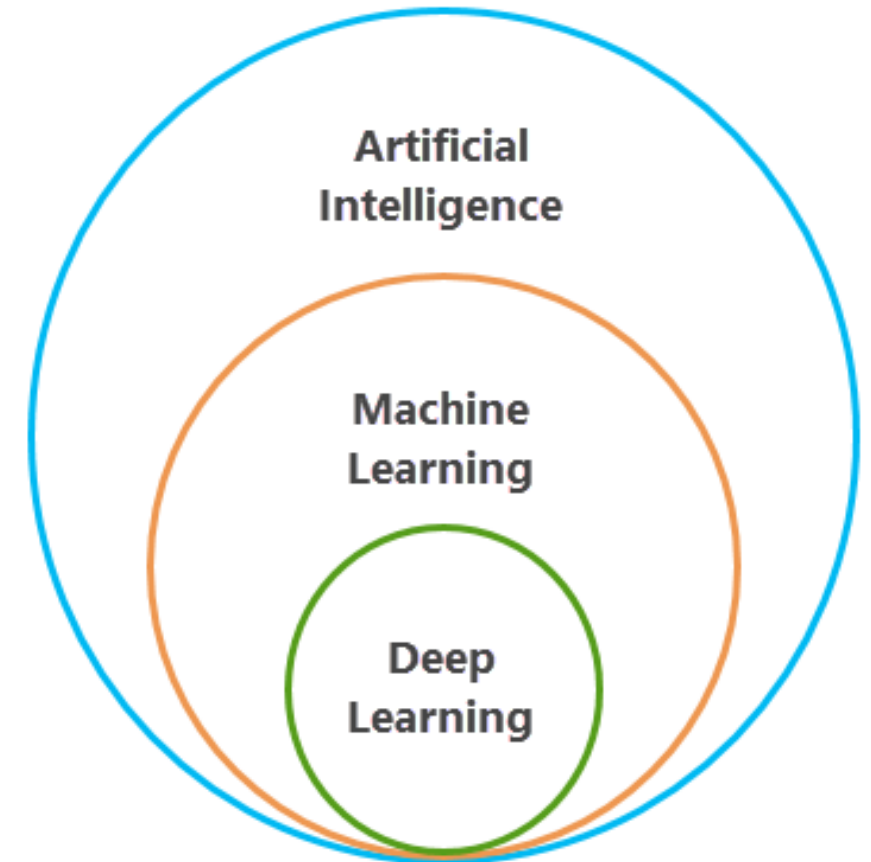


Deep Learning: Introducción

(CAPÍTULO 3 – CV03A)

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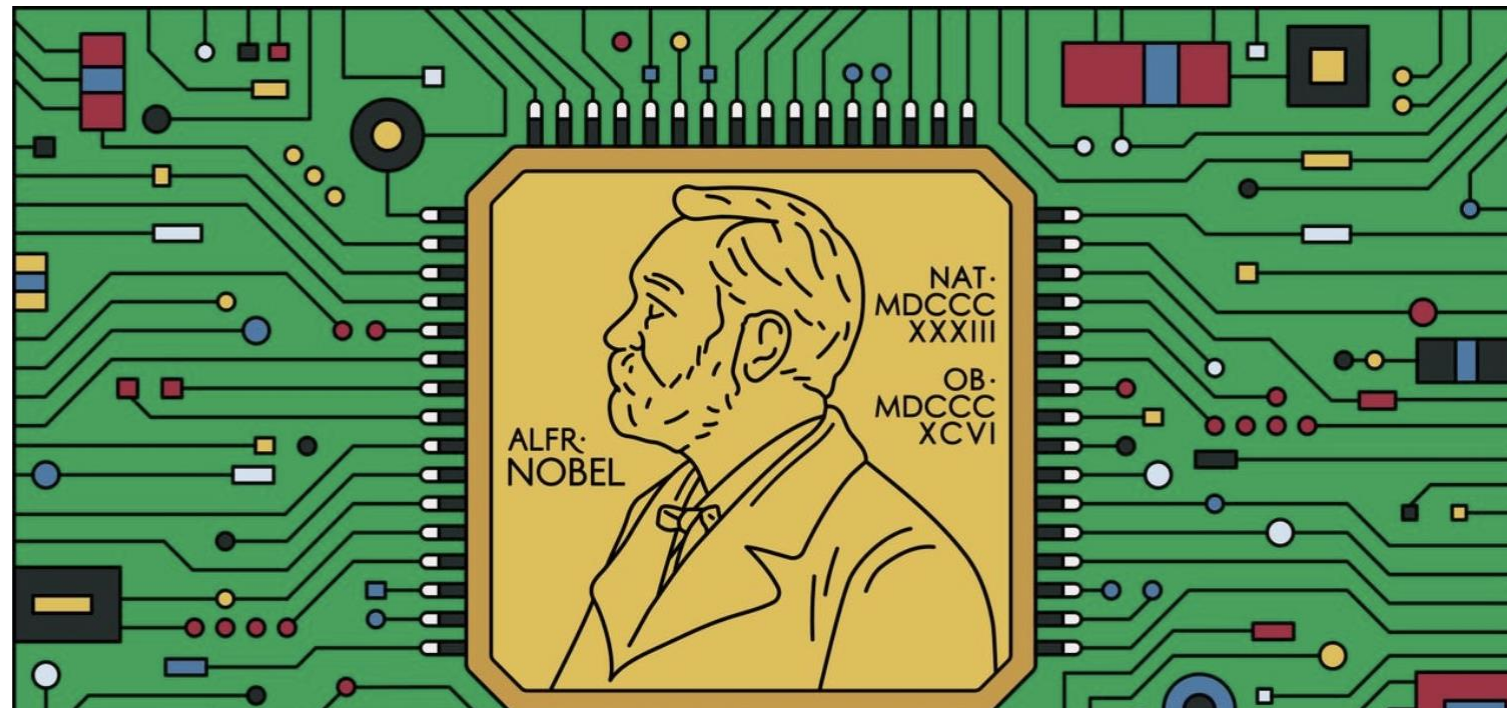
What If?

Jul 3rd 2021 edition >

Rage against the machine: December 2036

What if an AI won the Nobel prize for medicine?

Controversy ensues when the greatest prize in medical research is awarded to a non-human. An imagined scenario from 2036



ARTICLE OPEN

Development and validation of a deep learning algorithm for improving Gleason scoring of prostate cancer

Kunal Nagpal¹, Davis Foote¹, Yun Liu¹, Po-Hsuan Cameron Chen¹, Ellery Wulczyn¹, Fraser Tan¹, Niels Olson², Jenny L. Smith², Arash Mohtashamian², James H. Wren³, Greg S. Corrado¹, Robert MacDonald¹, Lily H. Peng¹, Mahul B. Amin⁴, Andrew J. Evans⁵, Ankur R. Sangoi⁶, Craig H. Mermel¹, Jason D. Hipp¹ and Martin C. Stumpe^{1,7}

For prostate cancer patients, the Gleason score is one of the most important prognostic factors, potentially determining treatment independent of the stage. However, Gleason scoring is based on subjective microscopic examination of tumor morphology and suffers from poor reproducibility. Here we present a deep learning system (DLS) for Gleason scoring whole-slide images of prostatectomies. Our system was developed using 112 million pathologist-annotated image patches from 1226 slides, and evaluated on an independent validation dataset of 331 slides. Compared to a reference standard provided by genitourinary pathology experts, the mean accuracy among 29 general pathologists was 0.61 on the validation set. The DLS achieved a significantly higher diagnostic accuracy of 0.70 ($p = 0.002$) and trended towards better patient risk stratification in correlations to clinical follow-up data. Our approach could improve the accuracy of Gleason scoring and subsequent therapy decisions, particularly where specialist expertise is unavailable. The DLS also goes beyond the current Gleason system to more finely characterize and quantitate tumor morphology, providing opportunities for refinement of the Gleason system itself.

npj Digital Medicine (2019)2:48; <https://doi.org/10.1038/s41746-019-0112-2>



A U-Net Deep Learning Framework for High Performance Vessel Segmentation in Patients With Cerebrovascular Disease

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¹Predictive Modelling in Medicine Research Group, Department of Neurosurgery, Charité - Universitätsmedizin Berlin, Berlin, Germany

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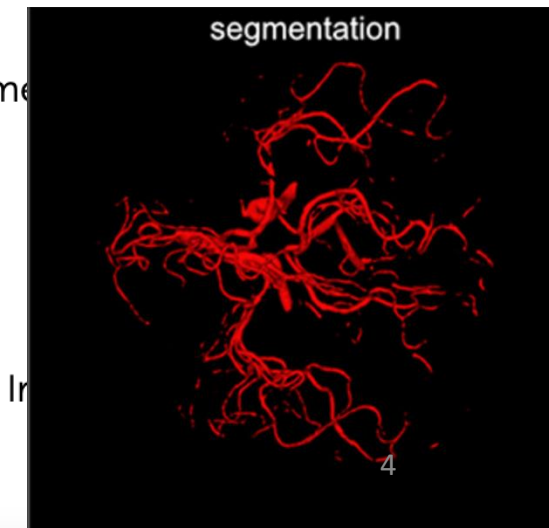
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⁴Department of Neurology, Johanna-Etienne Hospital Neuss, Neuss, Germany

⁵Information, Communication and Entertainment Institute (ICE), Dublin Institute of Technology, Dublin, Ireland

⁶Department VI Computer Science and Media, Beuth University of Applied Sciences, Berlin, Germany

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Dermatologist-level classification of skin cancer with deep neural networks

Andre Esteva^{1*}, Brett Kuprel^{1*}, Roberto A. Novoa^{2,3}, Justin Ko², Susan M. Swetter^{2,4}, Helen M. Blau⁵ & Sebastian Thrun⁶

Skin cancer, the most common human malignancy^{1–3}, is primarily diagnosed visually, beginning with an initial clinical screening and followed potentially by dermoscopic analysis, a biopsy and histopathological examination. Automated classification of skin lesions using images is a challenging task owing to the fine-grained variability in the appearance of skin lesions. Deep convolutional neural networks (CNNs)^{4,5} show potential for general and highly variable tasks across many fine-grained object categories^{6–11}. Here we demonstrate classification of skin lesions using a single CNN, trained end-to-end from images directly, using only pixels and disease labels as inputs. We train a CNN using a dataset of 129,450 clinical images—two orders of magnitude larger than previous datasets¹²—consisting of 2,032 different diseases. We

images (for example, smartphone images) exhibit variability in factors such as zoom, angle and lighting, making classification substantially more challenging^{23,24}. We overcome this challenge by using a data-driven approach—1.41 million pre-training and training images make classification robust to photographic variability. Many previous techniques require extensive preprocessing, lesion segmentation and extraction of domain-specific visual features before classification. By contrast, our system requires no hand-crafted features; it is trained end-to-end directly from image labels and raw pixels, with a single network for both photographic and dermoscopic images. The existing body of work uses small datasets of typically less than a thousand images of skin lesions^{16,18,19}, which, as a result, do not generalize well to new images. We demonstrate generalizable classification with a new

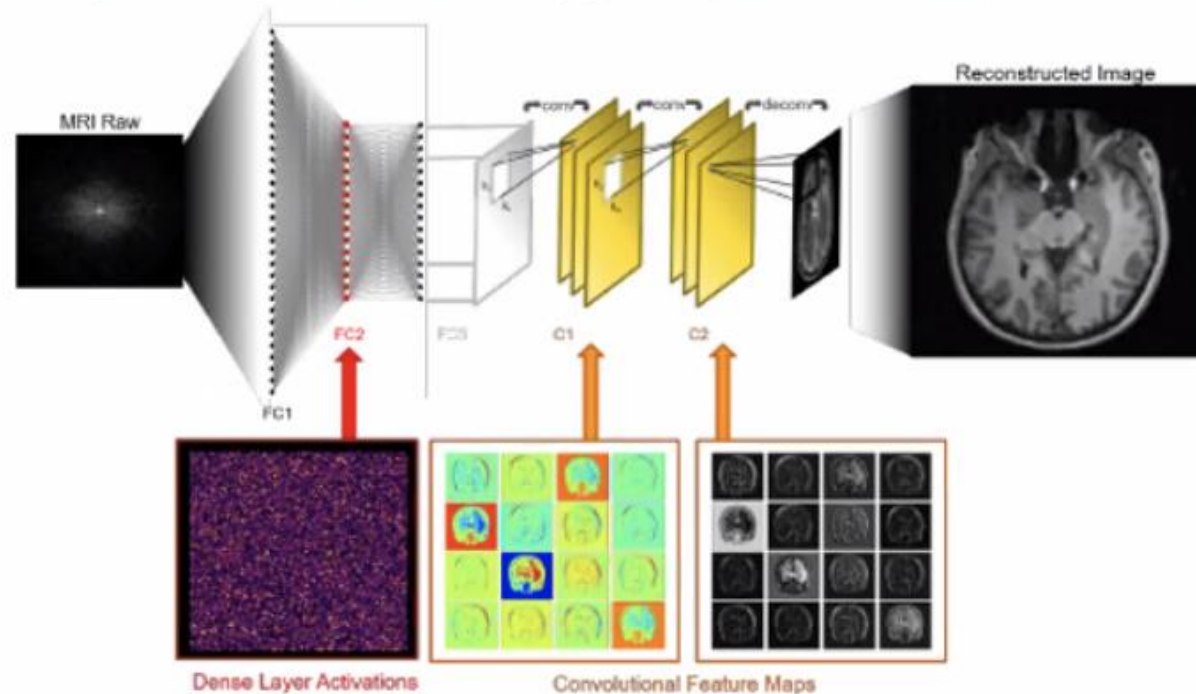
Image reconstruction by domain-transform manifold learning

Bo Zhu^{1,2,3}, Jeremiah Z. Liu⁴, Stephen F. Cauley^{1,2}, Bruce R. Rosen^{1,2} & Matthew S. Rosen^{1,2,3}

Image reconstruction is essential for imaging applications across the physical and life sciences, including optical and radar systems, magnetic resonance imaging, X-ray computed tomography, positron emission tomography, ultrasound imaging and radio astronomy^{1–3}. During image acquisition, the sensor encodes an intermediate representation of an object in the sensor domain, which is subsequently reconstructed into an image by an inversion of the encoding function. Image reconstruction is challenging

Inspired by the perceptual learning archetype, we describe here a data-driven unified image reconstruction approach, which we call AUTOMAP, that learns a reconstruction mapping between the sensor-domain data and image-domain output (Fig. 1a). As this mapping is trained, a low-dimensional joint manifold of the data in both domains is implicitly learned (Fig. 1b), capturing a highly expressive representation that is robust to noise and other input perturbations.

We implemented the AUTOMAP unified reconstruction framework





Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Review

Deep learning on image denoising: An overview

Chunwei Tian^{a,b}, Lunke Fei^c, Wenxian Zheng^d, Yong Xu^{a,b,e,*}, Wangmeng Zuo^{f,e},
Chia-Wen Lin^g

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ARTICLE INFO

Article history:

Received 16 January 2020

Received in revised form 17 June 2020

Accepted 21 July 2020

Available online 6 August 2020

Keywords:

Deep learning

Image denoising

Real noisy images

Blind denoising

Hybrid noisy images

ABSTRACT

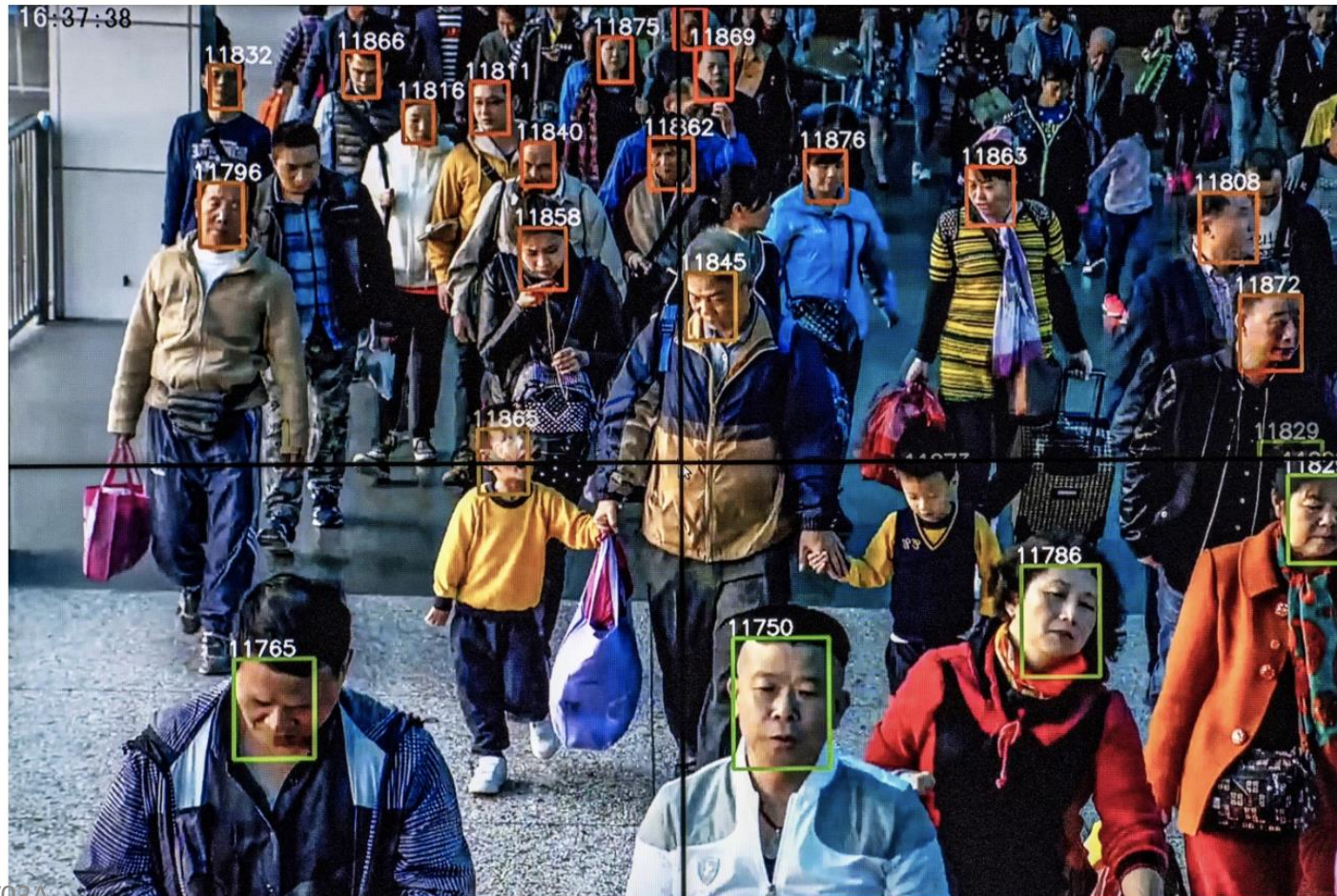
Deep learning techniques have received much attention in the area of image denoising. However, there are substantial differences in the various types of deep learning methods dealing with image denoising. Specifically, discriminative learning based on deep learning can ably address the issue of Gaussian noise. Optimization models based on deep learning are effective in estimating the real noise. However, there has thus far been little related research to summarize the different deep learning techniques for image denoising. In this paper, we offer a comparative study of deep techniques in image denoising. We first classify the deep convolutional neural networks (CNNs) for additive white noisy images; the deep CNNs for real noisy images; the deep CNNs for blind denoising and the deep CNNs for hybrid noisy images, which represents the combination of noisy, blurred and low-resolution images. Then, we analyze the motivations and principles of the different types of deep learning methods. Next, we compare the state-of-the-art methods on public denoising datasets in terms of quantitative and qualitative analyses. Finally, we point out some potential challenges and directions of future research.

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In China, Facial Recognition Tech Is Watching You

BY EAMON BARRETT

October 28, 2018 11:00 AM GMT-3



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Where's Waldo? Monitors at the Beijing offices of A.I.-software startup Megvii play a video showing how its facial recognition software

How Tesla Is Using Artificial Intelligence to Create The Autonomous Cars Of The Future

Over 500,000 Teslas all over the world are feeding data back to Elon Musk's headquarters, to train their autonomous car algorithms. This data gives Tesla a huge advantage in the race to put more self-driving cars on the road.



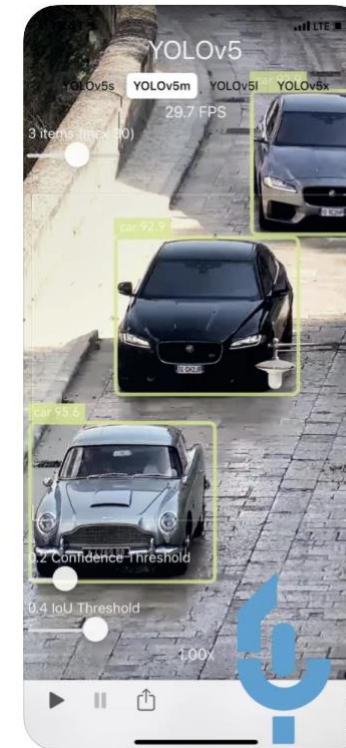
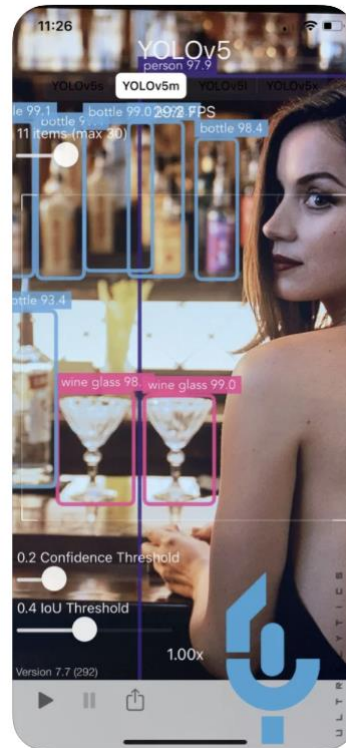
This app is available only on the App Store for iPhone and iPad.



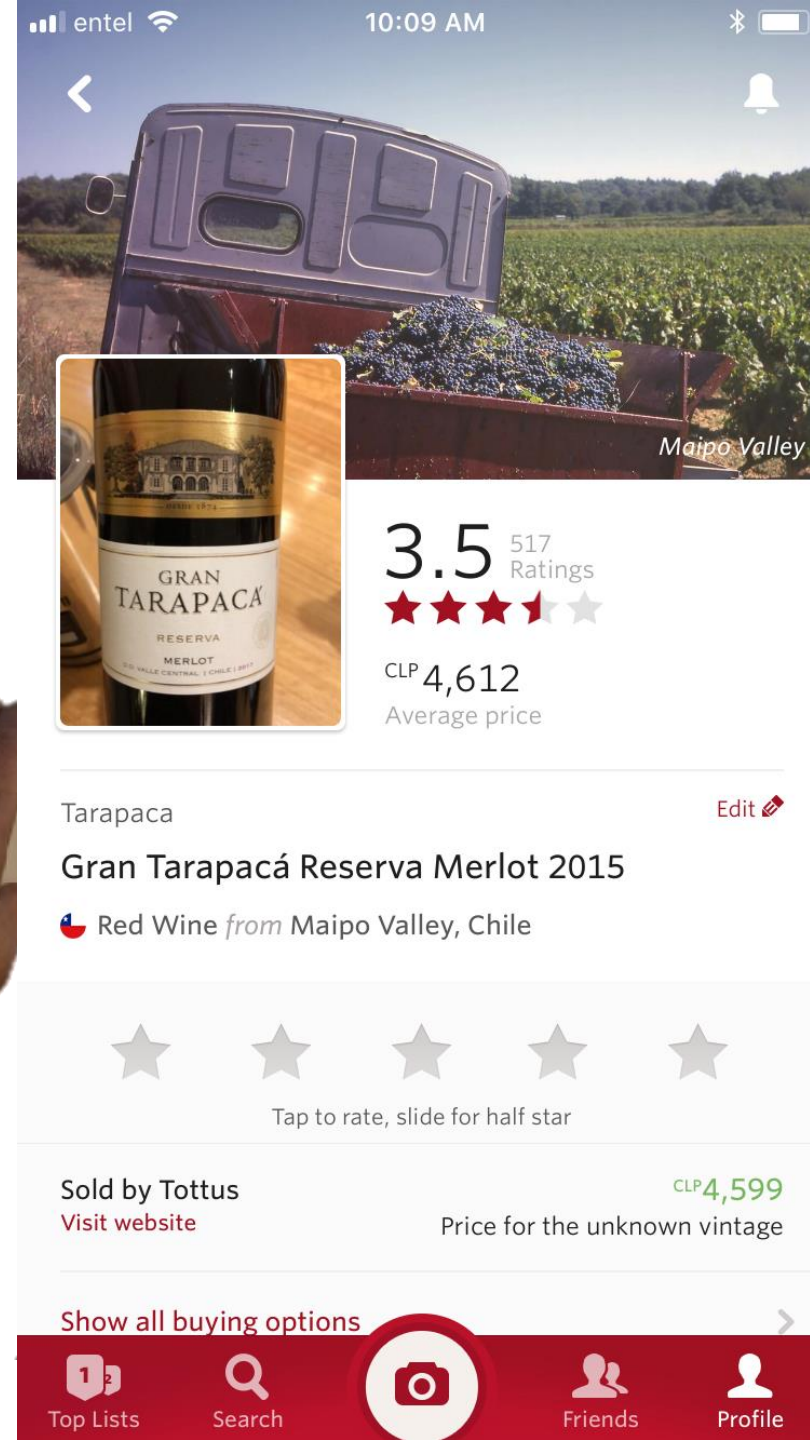
iDetection 4+
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[Ultralytics LLC](#)
★★★★★ 4.5 • 39 Ratings
Free

Object Detection

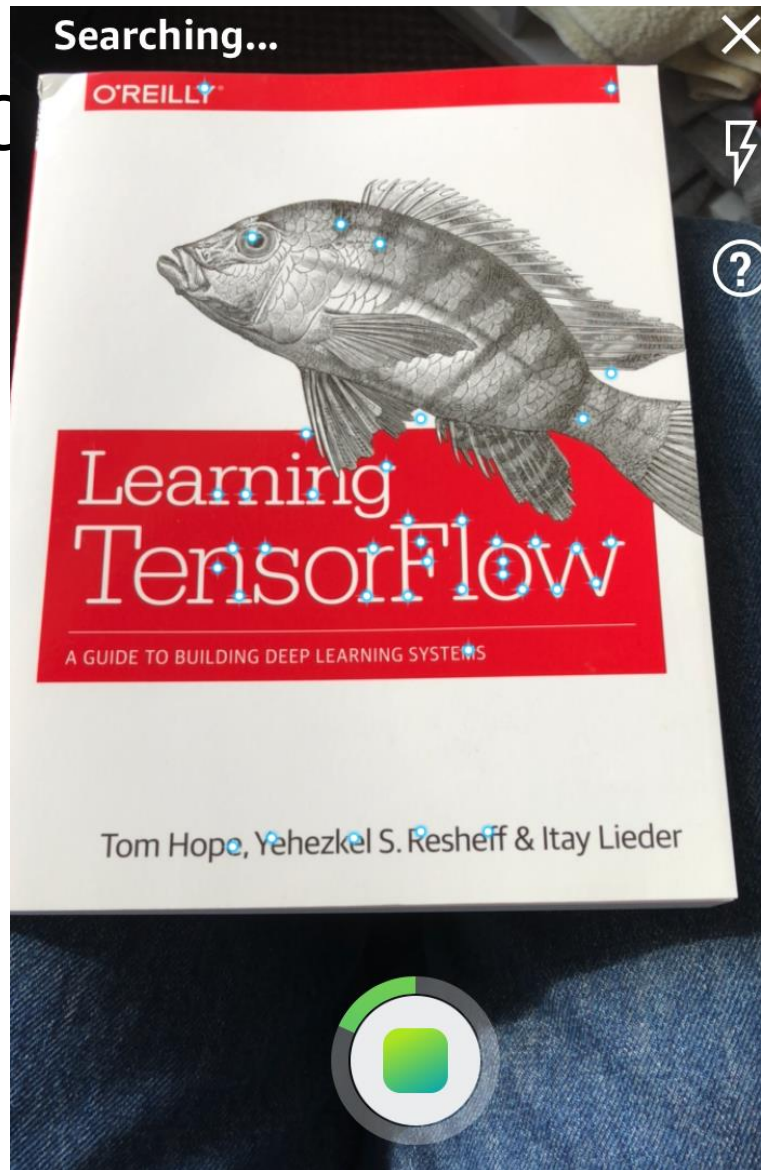
Screenshots [iPhone](#) [iPad](#)



Applications: Vivino



Applic



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Learning Node: Moving to the Server-Side



Powered by  Synthesia



106 STARTUPS TRANSFORMING HEALTHCARE WITH AI



... etcetera

Applications of Computer Vision

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> Five Applications of Computer Vision

Applications of Computer Vision

- 1) Image Classification
- 2) Image Processing
- 3) Object Detection
- 4) Anomaly Detection
- 5) Human + Computer Vision

- Problem
- Solution
- Examples

Applications of Computer Vision

Applications

- 1) Image Classification
- 2) Image Processing
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Applications of Computer Vision

Applications

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Image Classification – Problem

> Assign an image to a class or category

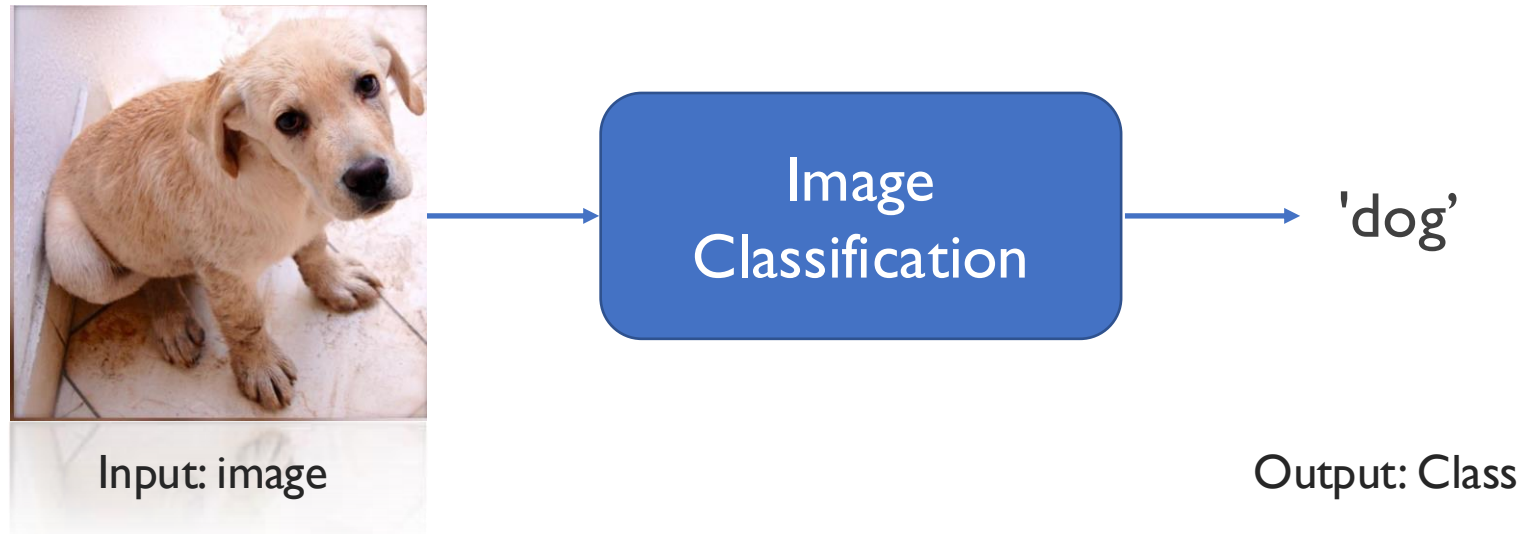


Image Classification – Problem

> Assign an image to a class or category

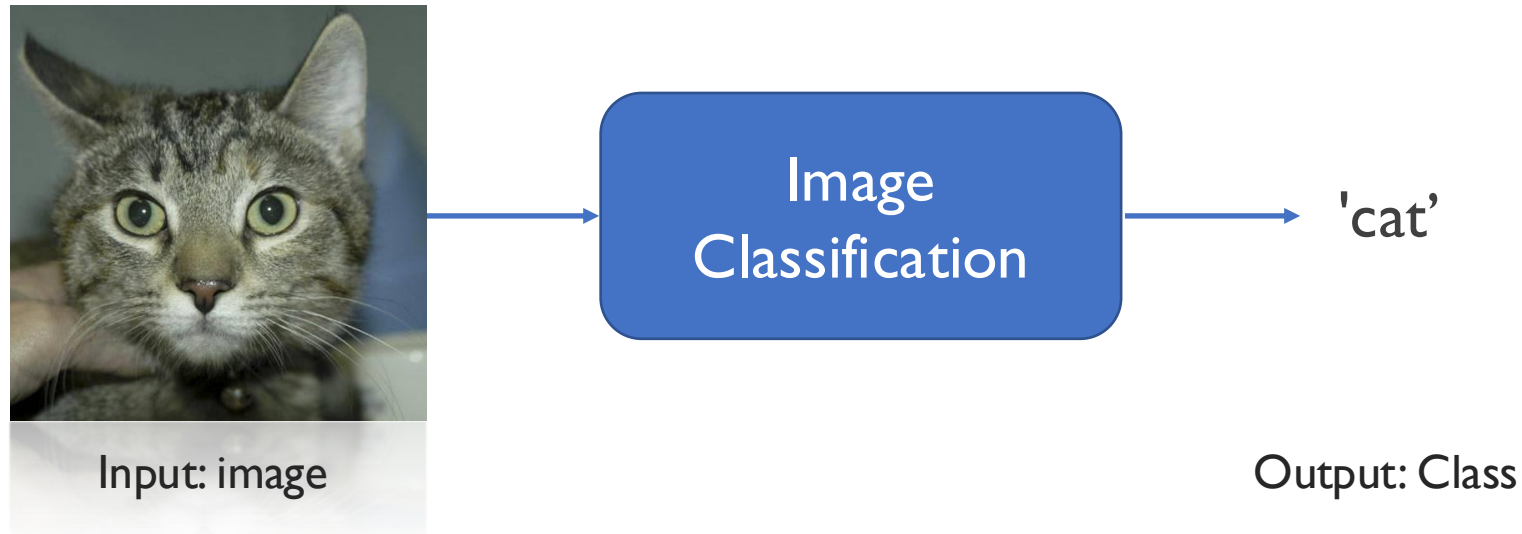
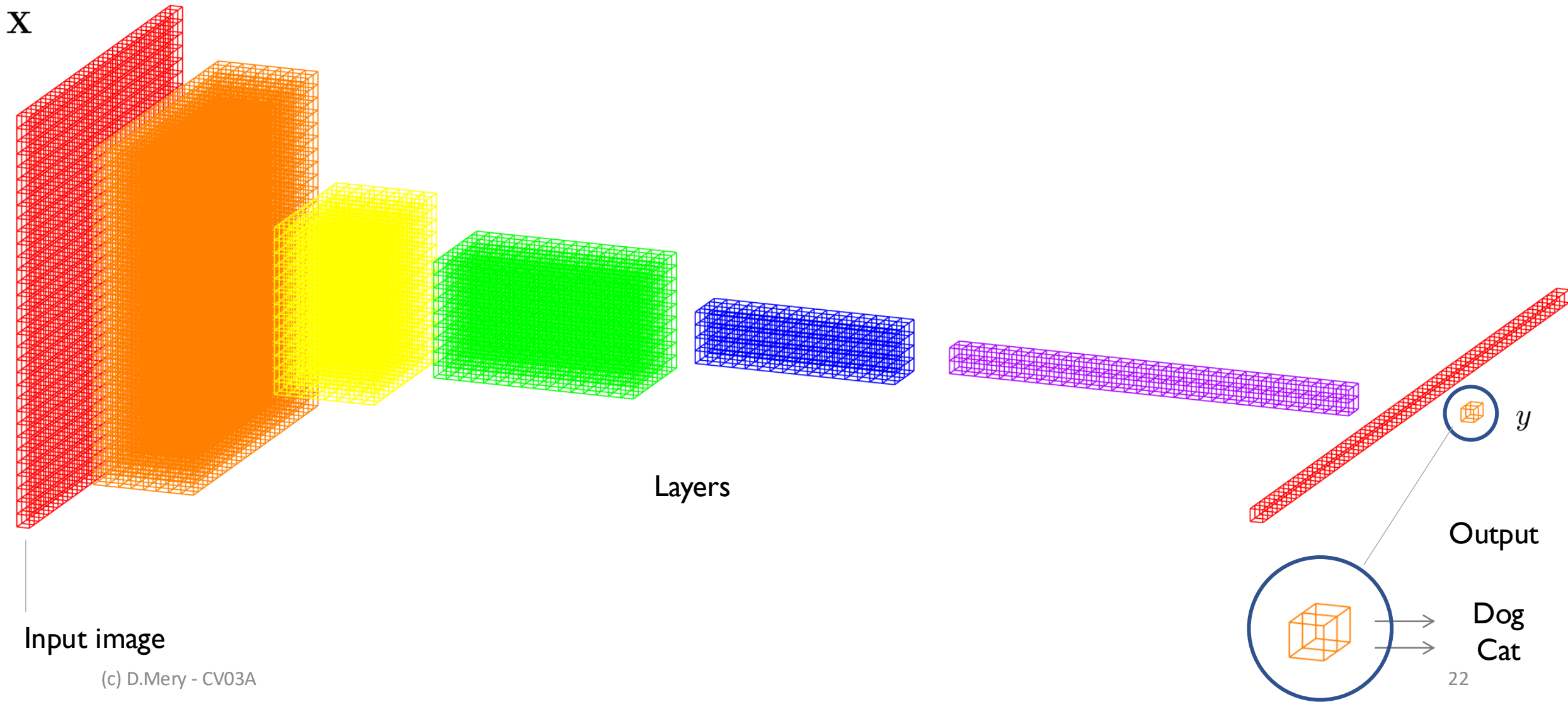


Image Classification – Solution

Deep Learning Model: CNN Convolutional Neural Network



Applications of Computer Vision

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Image Processing – Problem

> Get a target output image from an input image

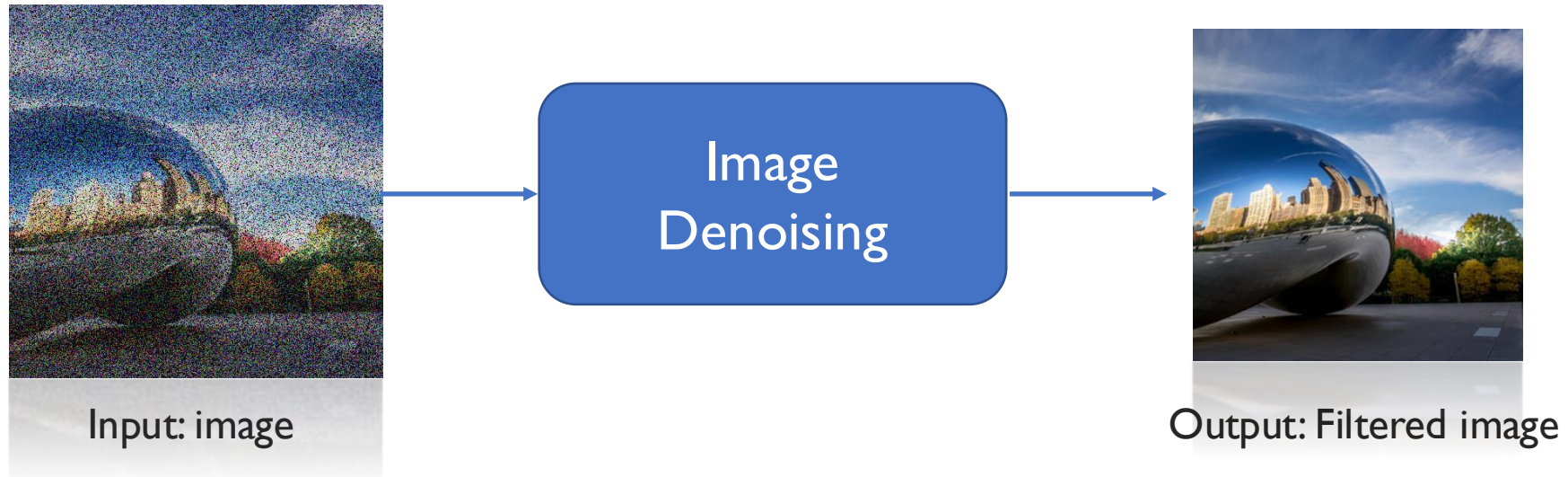


Image Processing – Problem

> Get a target output image from an input image



Image Processing – Solution

Deep Learning Model: U-Net

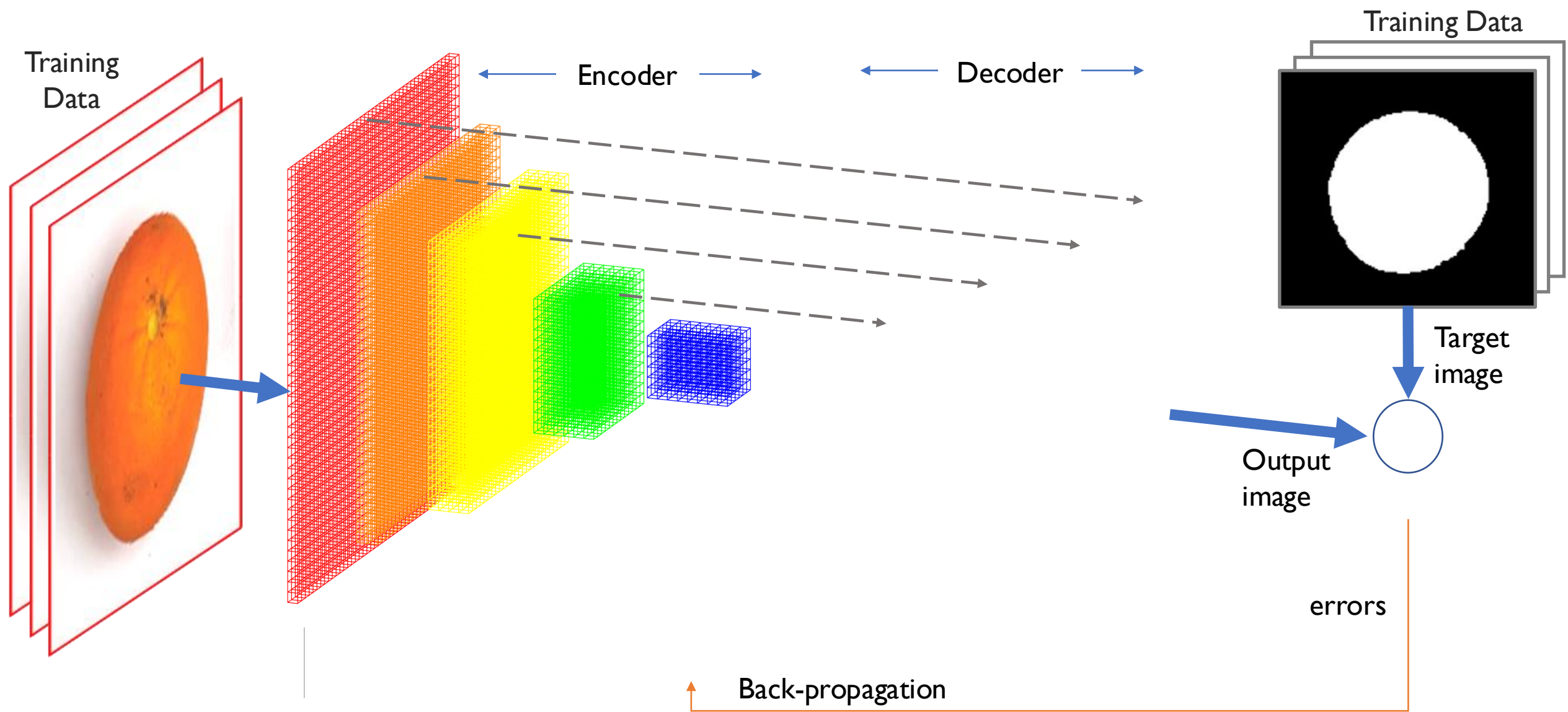
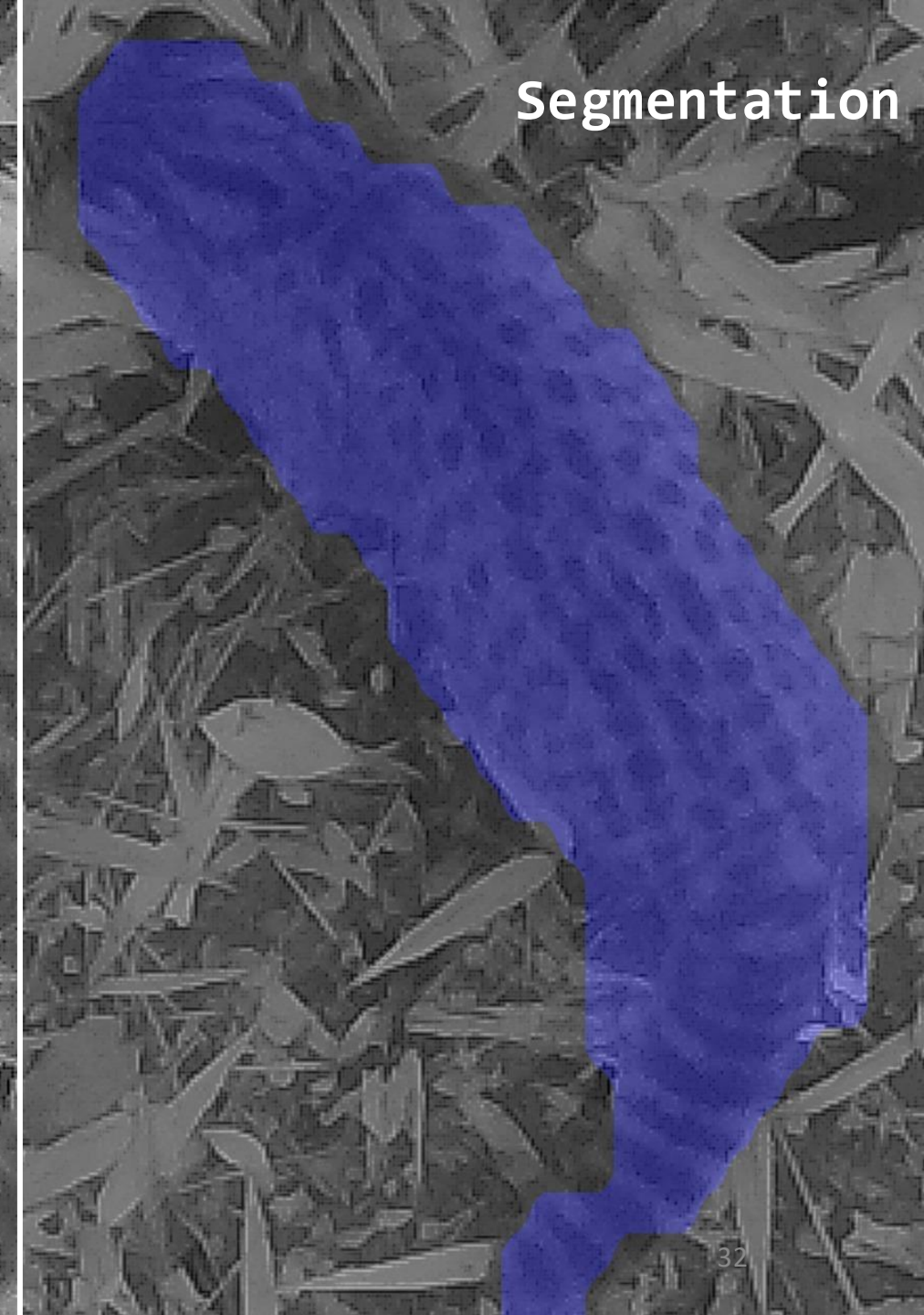


Image Segmentation Example

Güiñas Segmentation



Applications of Computer Vision for Medicine

Applications

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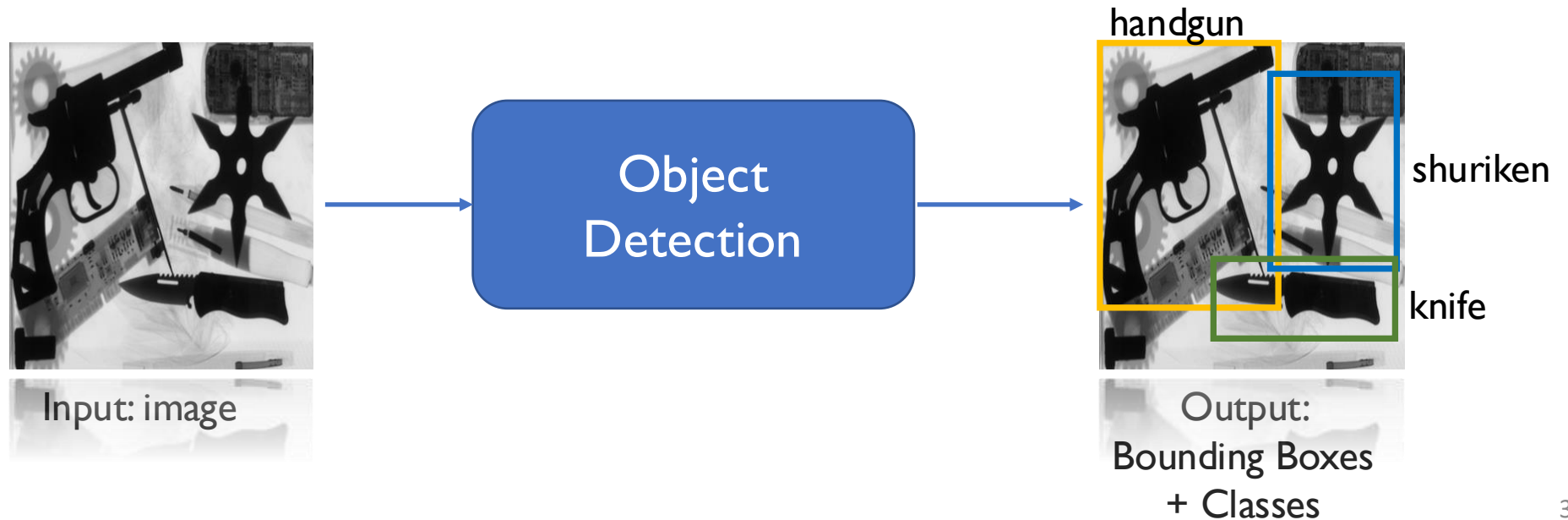
Applications of Computer Vision

Applications

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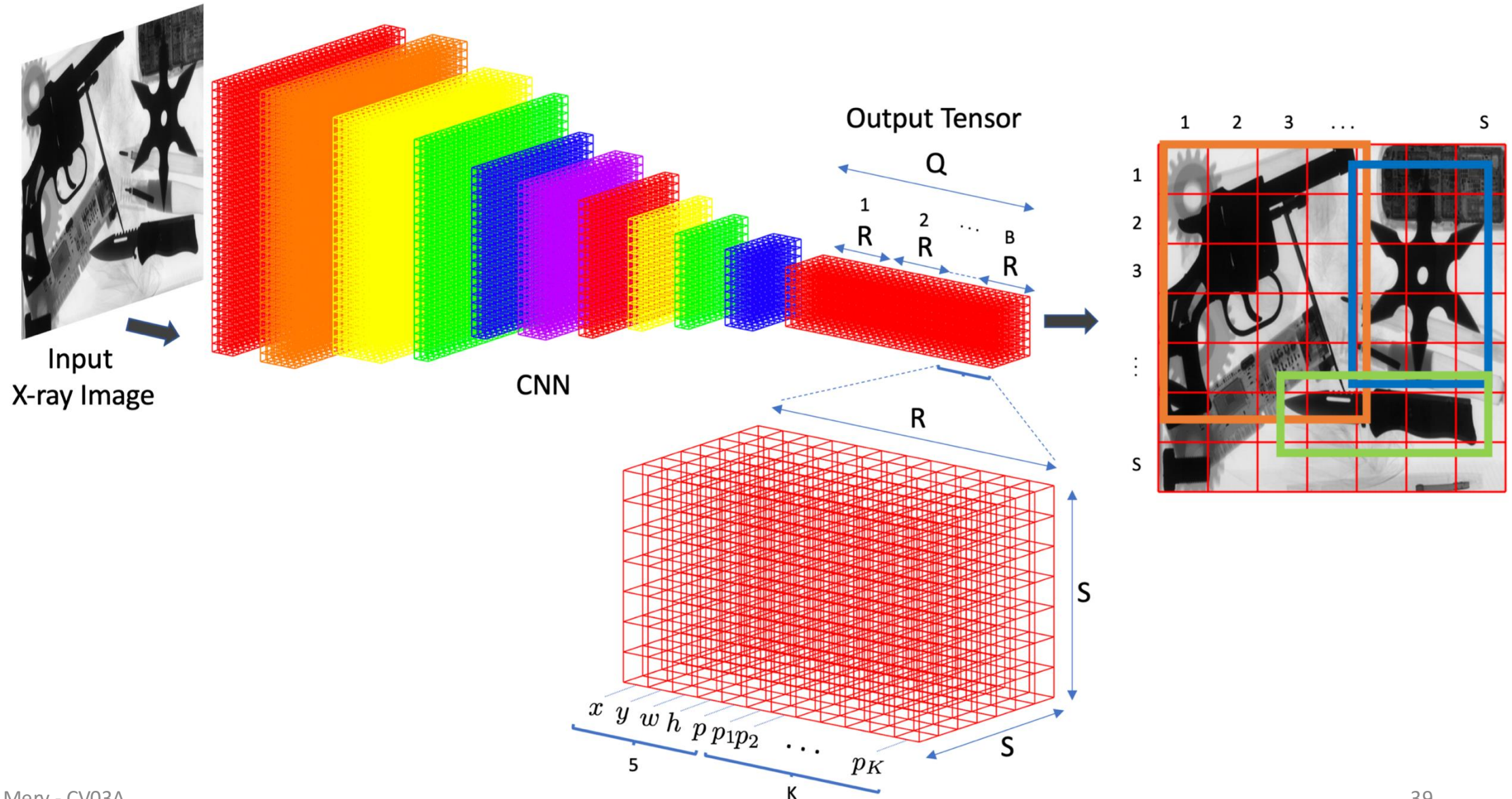
Object Detection – Problem

> Locate and classify objects



Object Detection – Solution

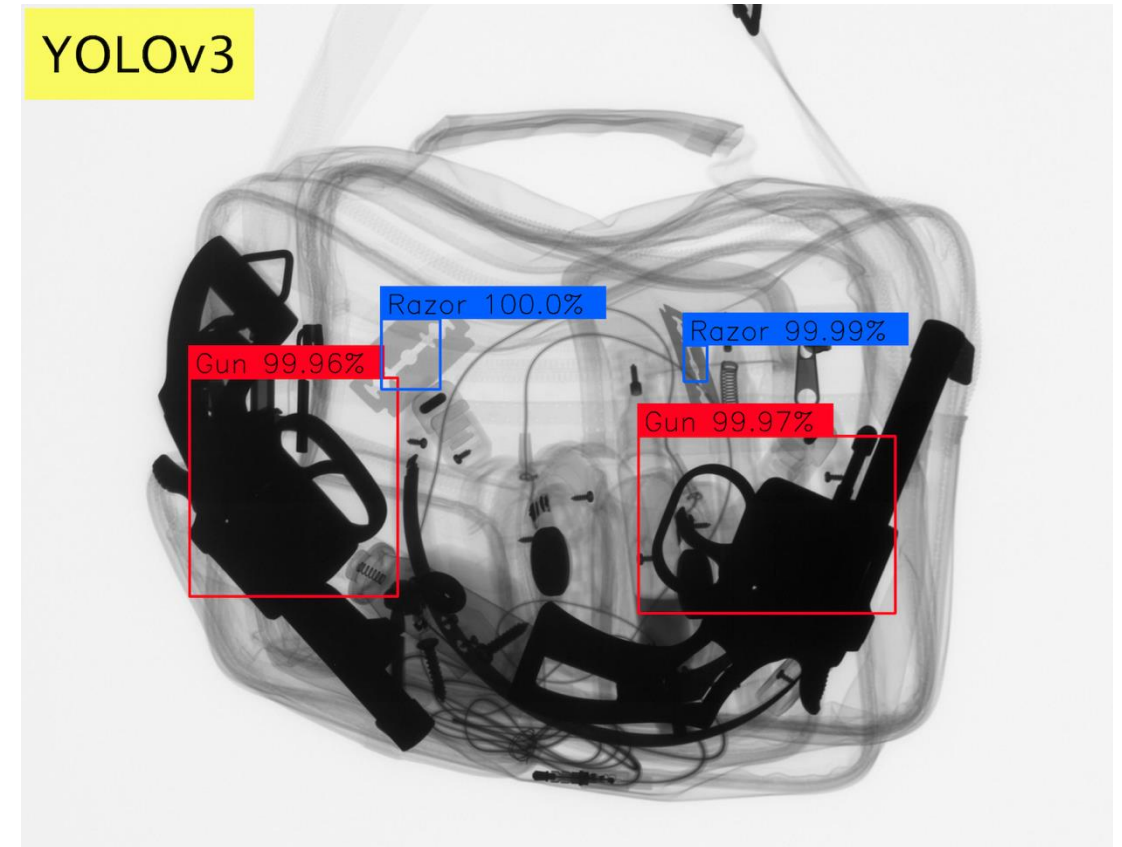
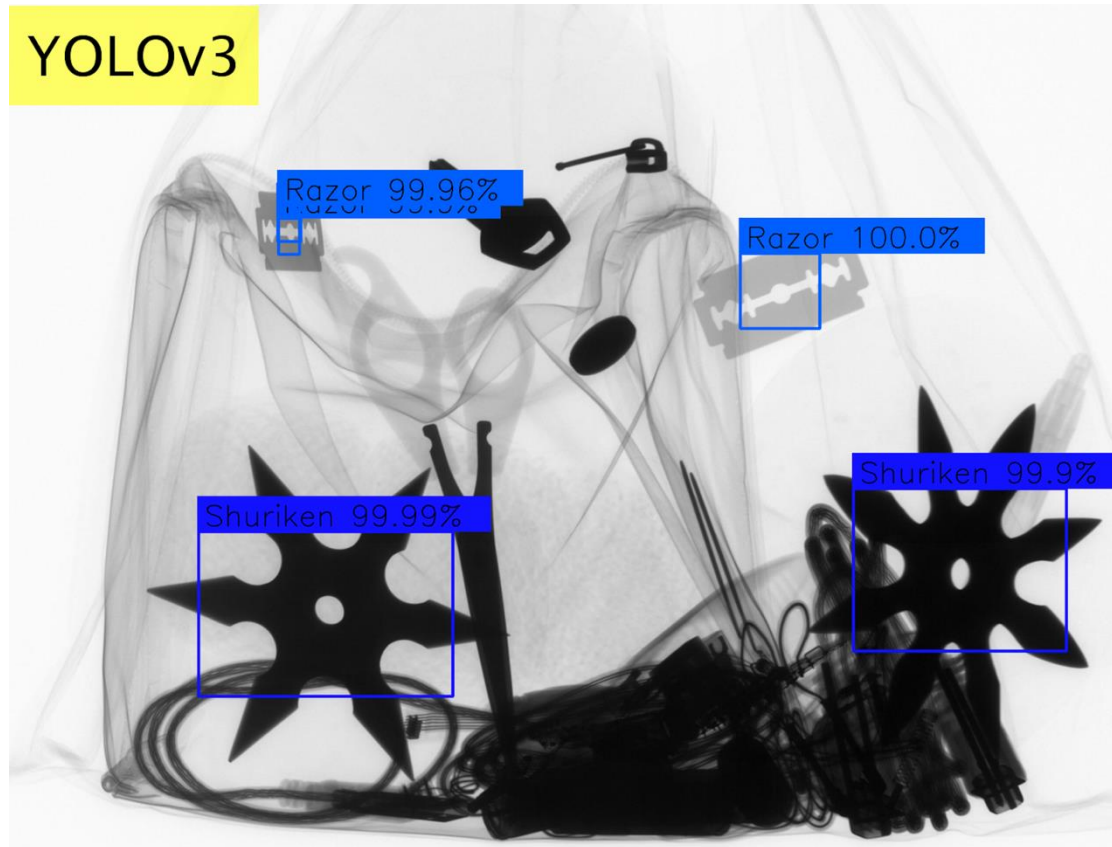
Deep Learning Model:YOLO



Object Detection – Example

Deep Learning Model:YOLO

Recognition of Threat Objects in Baggage Inspection



Applications of Computer Vision

Applications

- 1) Image Classification
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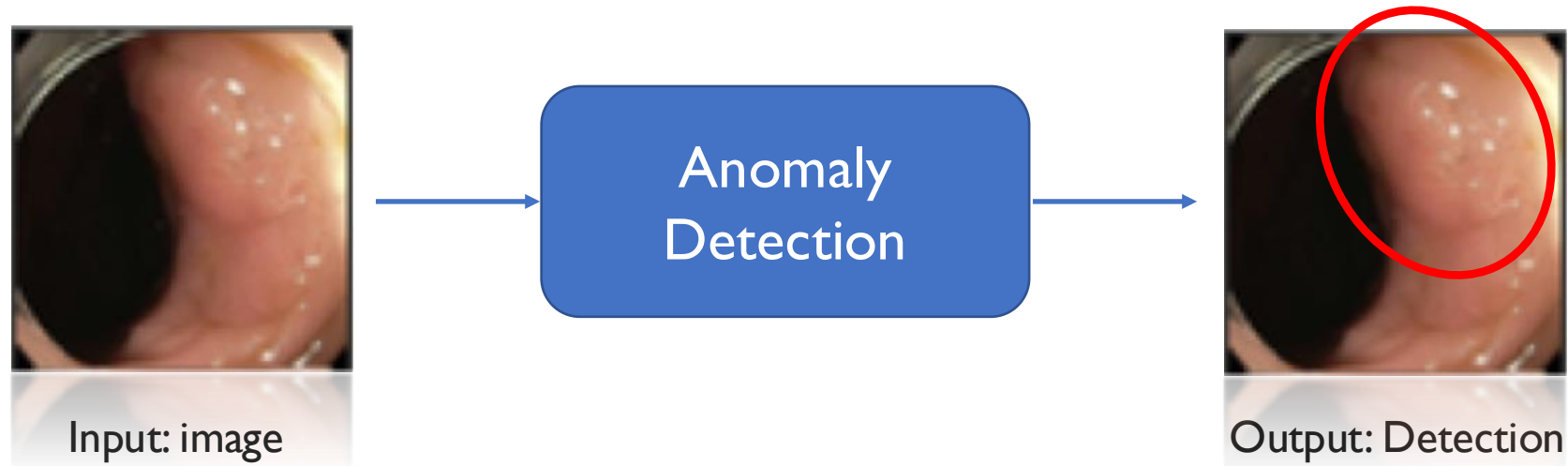
Applications of Computer Vision

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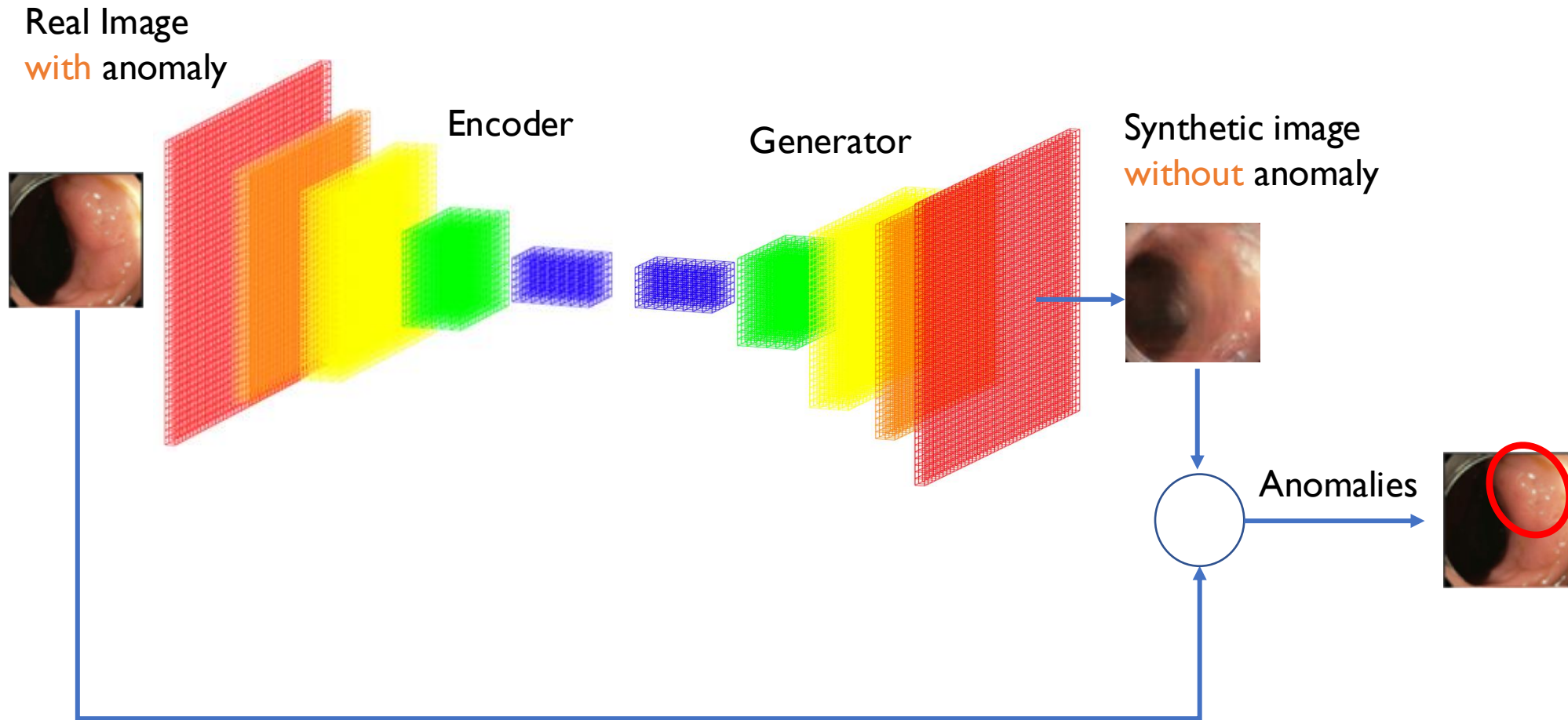
Anomaly Detection – Problem

> Detect an anomaly in an image



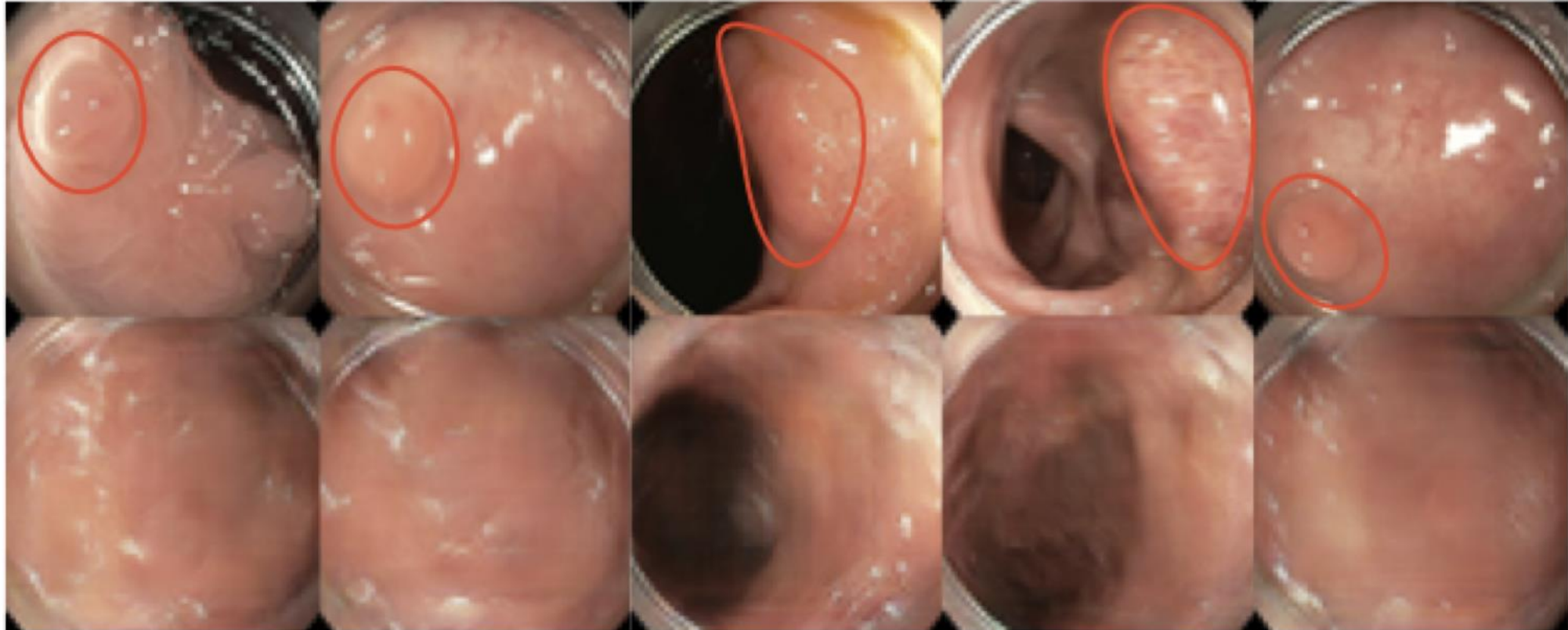
Anomaly Detection – Solution

Deep Learning Model: GAN Generative Adversarial Network



Anomaly Detection – Example

Colonoscopy video frames



Liu, Yuyuan, et al. "Photoshopping colonoscopy video frames." *2020 IEEE 17th International Symposium on Biomedical Imaging (ISBI)*. IEEE, 2020.

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Applications of Computer Vision

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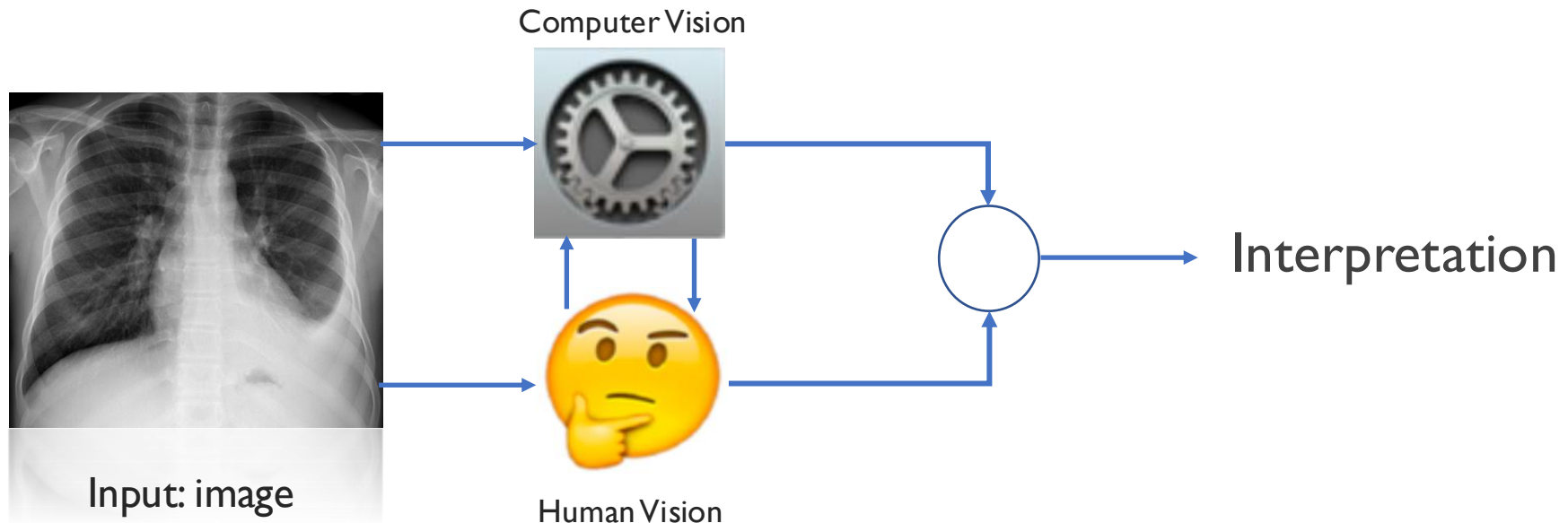
Applications of Computer Vision

Applications

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Human + Computer Vision – Problem

> How can Human Vision and Computer Vision work in collaboration?



Applications of Computer Vision

Applications

- 1) Image Classification
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Applications of Computer Vision

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Algoritmos de Deep Learning del Curso

- Object Classification (CNN)
- Object Detection (YOLO)
- Tracking
- Segmentation (UNet)
- GAN
- Anomaly Detection
- Transformers
- Facial Analysis

- Face detection
- Face recognition
- Face restoration
- Age recognition
- Gender recognition
- Expression recognition
- Soft attributes
- Face clustering
- Pose estimation
- Explainability